

GARUDA

(Web application)

COMPLETE DPR FORMAT

By,
Tirupati District Police

COMPLETE DPR FORMAT

NDPS ACCUSED MONITORING & INTELLIGENCE MANAGEMENT WEB APPLICATION

DETAILED PROJECT REPORT (DPR)

1.Cover Page

Title: GARUDA

Detailed Project Report (DPR) For NDPS Accused Monitoring & Intelligence Management Web Application by Tirupati district Police department.

2.Executive Summary :-

This project proposes development of a centralized secure web-based platform for monitoring, surveillance, intelligence gathering, and analysis of NDPS accused persons, habitual offenders, interstate narcotics traffickers, and organized drug networks.

The system will enable:

- Centralized accused database
- Intelligence sharing
- Surveillance monitoring
- Link analysis
- GIS crime mapping
- Real-time reporting
- AI-based analytics (future scope)

The application will strengthen intelligence-led policing and improve anti-narcotics enforcement efficiency.

3. Introduction :-

Drug trafficking and narcotics-related crimes have emerged as serious threats affecting public safety, youth, and law enforcement. Presently, offender-related information is fragmented across police stations and units.

To address this issue, a centralized NDPS accused monitoring web application is proposed for:

- Criminal intelligence management
- Habitual offender tracking
- Interstate coordination
- Surveillance operations
- Data-driven enforcement

4. Objectives

a.Primary Objectives

- Centralized digital database of NDPS accused
- Real-time intelligence sharing
- Monitoring repeat offenders
- Enhancing surveillance operations
- Data analytics for enforcement

b.Secondary Objectives

- GIS hotspot mapping
- Link analysis
- Crime pattern analysis
- Automated alerts
- AI-based predictive policing

5. Existing System Study

Existing Problems

- Manual record maintenance
- Fragmented data
- No centralized database
- Difficulty tracking interstate offenders
- Lack of analytical tools
- Delay in intelligence dissemination
- Limited offender profiling

6. Proposed System

The proposed web application shall provide:

- Centralized offender database
- Surveillance management
- Mobile-based field updates
- Network analysis
- GIS integration
- Real-time dashboards
- Advanced search mechanisms

7. Scope of the Project

The system shall cover:

- All NDPS accused
- Suspected peddlers
- Repeat offenders
- Habitual traffickers
- Organized drug syndicates
- Interstate narcotics operators

Coverage:

- Police Stations
- SEB Units
- Narcotics Cells
- Intelligence Units
- District & State Headquarters

8. Functional Requirements

8.1 User Management

- Login system
- Password policy
- Role-based access
- User creation/modification
- Access logs

8.2 Accused Database Module

Fields:

- Personal details
- Photograph
- Biometrics
- Address
- Mobile numbers
- Vehicle details
- Social media accounts
- Associates
- Criminal history

8.3 Case Management Module

- FIR details
- Arrest information
- Seizure details
- Charge sheet tracking
- Court case status
- Bail details

8.4 Surveillance Module

- History sheets
- Monitoring schedules
- Field verification reports
- Activity tracking
- Geo-tagging

8.5 Intelligence Module

- Source inputs
- Confidential notes
- Network intelligence
- Supply route analysis

8.6 Search & Analytics

Search Types:

- Name
- Mobile
- Vehicle
- FIR
- Aadhaar
- District-wise

Analytics:

- Repeat offender analysis
- Drug trend analysis
- Hotspot analysis

8.7 Alerts & Notifications

- Bail alerts
- Repeat offender alerts
- Interstate movement alerts
- Surveillance due alerts

9. Non-Functional Requirements

a. Performance

- Fast response time
- Concurrent users- support

b. Scalability

- Expandable statewide deployment

c. Reliability

- 99.9% uptime

d. Availability

- 24x7 access

e. Security

- Encryption
- MFA
- Audit logs

10. Technical Architecture

a. Frontend

- React.js
- HTML5
- CSS3
- JavaScript

b. Backend

- Node.js / Django

c. Database

- PostgreSQL

d. Hosting+

- NIC Cloud / State Data Center

11. System Architecture Diagram

a. Components

- User Interface Layer
- Application Layer
- Database Layer
- Analytics Engine
- GIS Layer
- Security Layer

(Insert Architecture Diagram)

12. Database Design

a. Main Tables

- Users
- Accused
- Cases
- Surveillance
- Vehicles
- Associates
- Alerts
- Intelligence Inputs

(Insert ER Diagram)

13. Security Requirements

Cyber Security Features

- HTTPS
- SSL/TLS
- Role-based access
- Data encryption
- IP restriction
- Session timeout
- Audit trails
- VAPT
- SIEM integration

14. Infrastructure Requirements

a. Hardware

- Application server
- Database server
- Backup server

b. Network

- Firewall
- VPN
- IDS/IPS

c. Storage

- RAID storage
- Backup NAS

15. Integration Requirements

Proposed Integrations

- CCTNS
- ICJS
- CCTV systems
- GIS maps
- Prison databases
- Vehicle databases
- Facial recognition

16. GIS & Mapping Features

- Hotspot mapping
- Route analysis
- Drug corridor mapping
- Heat maps
- Vulnerable area mapping

17. AI-Based Future Scope

- Facial recognition
- Predictive analytics
- Pattern detection
- Automated risk scoring
- Network anomaly detection

18. Implementation Methodology

a.Phase 1

Requirement Gathering

b.Phase 2

System Design

c.Phase 3

Development

d.Phase 4

Testing

e.Phase 5

Deployment

f.Phase 6

Training

g.Phase 7

Maintenance

19. Testing Plan

a.Testing Types

- Unit testing
- Integration testing
- Performance testing
- Load testing
- UAT
- Security testing
- VAPT

20. Training Plan

a. Training For

- SHOs
- IOs
- Intelligence staff
- Technical staff

b.Materials

- SOPs
- Manuals
- Video tutorials

21. Maintenance & Support

a.Services

- AMC support
- Security patching
- Database optimization
- Technical helpdesk

22. Risk Assessment

Sl. No.	Risk	Mitigation
1	Data theft	Encryption
2	Server crash	DR site
3	Unauthorized access	MFA
4	Data corruption	Backup system

23. Budget Estimation

Sl.No.	Component	Estimated cost
1	Software Development	
2	Servers	
3	Security Audit	
4	Training	
5	AMC	

24. Expected Outcomes

- Improved intelligence collection
- Faster offender identification
- Better interstate coordination
- Stronger surveillance
- Data-driven enforcement
- Reduced repeat offences

25. Conclusion

The proposed system will significantly strengthen anti-narcotics enforcement capabilities through centralized intelligence management, surveillance monitoring, analytics, and technology-driven policing.

SOFTWARE REQUIREMENT SPECIFICATION (SRS) TEMPLATE

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Name: NDPS Accused Monitoring & Intelligence Management System

1. Introduction

1.1 Purpose :- Define purpose of the application.

1.2 Scope:- Define system scope.

1.3 Definitions:- List abbreviations and definitions.

Example:

- NDPS
- FIR
- GIS
- RBAC

2. Overall Description

2.1 Product Perspective :- Describe system overview.

2.2 Product Functions :- List major functionalities.

2.3 User Classes

- Admin
- SHO
- IO
- Intelligence Officer

2.4 Operating Environment

- Web browser
- Android support
- Cloud hosting

3. Functional Requirements

- FR-1 Login Module

3.1. Description:

3.2. Secure login.

a.Inputs:

3.3. Username/password

a. Outputs:

3.4. Dashboard access

FR-2 Accused Management

Features:

- Add accused
- Modify accused
- Search accused
- Delete accused

3.4. FR-3 Case Management

- FIR entry
- Charge sheet tracking
- Court updates

3.5. FR-4 Surveillance Management

- Verification scheduling
- Geo-tagging
- Monitoring reports

3.6. FR-5 Analytics Module

- Dashboard charts
- Crime trends
- Repeat offender analysis

4. Non-Functional Requirements

4.1. Security

- Encryption
- MFA
- Audit logs

4.2. Performance

- Response < 3 sec

4.3. Availability

- 24x7 uptime

4.4. Scalability

- State wide expansion

5. External Interface Requirements

5.1. User Interface

- Dashboard
- Search
- Reports

5.2. Hardware Interface

- Servers
- Biometric devices

5.3. Software Interface

- CCTNS
- GIS APIs

6. Database Requirements

6.1. Tables

- Users
- Accused
- FIR
- Vehicles
- Alerts

7. Security Requirements

- SSL
- RBAC
- VAPT
- Backup
- Audit trail

8. Reporting Requirements

8.1. Reports

- District-wise NDPS cases
- Repeat offenders
- Surveillance pending
- Bail monitoring

9. Constraints

- Government hosting compliance
- Cyber security compliance
- Limited internet connectivity in remote areas

10. Assumptions

- Users trained properly
- Secure network available

11. Future Enhancements

- AI analytics
- Facial recognition
- Mobile app
- Drone integration

12. Appendices

- ER diagrams
- Architecture diagrams
- SOPs
- Workflow charts